MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
QA/QC & INSPECTION CELL — PIPING FABRICATION

HYDROSTATIC PRESSURE TEST CERTIFICATE & CLEARANCE

CERTIFICATE NO: **MRPL-QC-HT-2026-042** | TEST DATE: 04-MAR-2026 | RESULT: **PASSED**

1. TEST PACKAGE SPECIFICATION

Piping Line Number:	2"-HC-1015-A1A (Minimum Flow Bypass Line)	P&ID Reference:	MRPL-CDU-PID-001 Rev 4
Design Pressure:	12.0 bar (g)	Calculated Test Pressure:	18.0 bar (g) [1.5 × P_des]
Test Medium:	DM Water with 50 ppm Hydrazine inhibitor	Ambient Temp / Fluid Temp:	31 °C / 29 °C
Holding Time:	4 Hours 00 Mins (10:00 to 14:00 IST)	Pressure Gauges Used:	PG-HT-01 & PG-HT-02 (Dual calibrated)

2. TEST LOG & WITNESS RECORD

Step	Time (IST)	Target Pressure	Observed Pressure	Hold Period	Visual Examination
Stage 1	10:00	6.0 bar (g)	6.05 bar (g)	15 mins	Initial leak check: No weep
Stage 2	10:30	12.0 bar (g)	12.00 bar (g)	15 mins	Check all flanged joints: No leak
Stage 3 (Full)	11:00	18.0 bar (g)	18.10 bar (g)	Begin 4h hold	Full circumference inspection OK
Final Reading	14:00	18.0 bar (g)	18.05 bar (g)	Completed	ZERO pressure drop; No deformation

3. INSPECTOR CONCURRENCE & PUNCH-LIST SIGN-OFF

Piping spool 2"-HC-1015-A1A successfully withstood the hydrotest pressure of 18.0 bar (g). Depressurization completed in controlled stages. Dewatering and air drying completed to dewpoint -20°C. Line cleared for reinstatement and final tie-in.

Quality Control Inspector	Third Party Inspector (EIL)	MRPL Operations Head
N. B. Rao (QC Lead)	T. S. Venkat (Engineers India Ltd)	P. K. Verma (DGM Operations)
[PASSED & SIGNED]	[VERIFIED & STAMPED]	[ACCEPTED FOR RESTORATION]